

Dear Vietnam Summer School of Science 2026's Selection Committee,

For me, **"Rethinking Science"** is not only about discovering new scientific ideas, but also about re-examining how a young researcher chooses problems, builds intellectual discipline, and learns to pursue science with integrity and purpose. Over the past year, this theme has become deeply personal to me.

During the later years of my undergraduate study at Vietnam National University, Hanoi, I gradually realized that enthusiasm alone was not enough for long-term research. Although I had been involved in robotics, control systems, and AI projects, I often questioned whether I truly understood the kind of scientific problems I wanted to pursue. I saw a gap between my curiosity and the intellectual maturity required to become a serious researcher. That realization led me to take a one-year gap period to step outside my familiar environment and comfort zone, strengthen my foundations, and rethink my academic direction before entering the final stage of my undergraduate journey.

During this period, I had the opportunity to join research programs in Taiwan and Singapore. At AIM-HI Institute, National Chung Cheng University, I worked on data-driven modeling for cutting force prediction in CNC machining. This experience helped me understand how AI and data are transforming industrial systems, but it also showed me that meaningful engineering research requires more than applying models: it requires understanding the foundation of the physical system, the limitations of data, and the assumptions behind each method.

Later, through the Global Connect Fellowship at Nanyang Technological University, Singapore, I worked on wireless power transfer for autonomous underwater vehicles. Although wireless power transfer was initially new to me, the project opened my view of how interdisciplinary modern research can be. A single problem required me to think across power electronics, mechanical design, control, and autonomous systems. More importantly, it helped me identify the research direction I genuinely want to pursue in my graduation thesis and future higher studies: intelligent autonomous systems, especially at the intersection of robotics, control, AI, and energy systems.

Looking back, these experiences were not simply academic opportunities. They taught me to rethink science as a disciplined process of asking better questions, accepting uncertainty, and continuously refining one's understanding. They also helped me reflect on my earlier attempts to apply to VSSS. Having applied to previous seasons without being selected, I now see those attempts not as failures, but as early signs of a strong motivation that has become more mature and better grounded through real research experience.

This is why VSSS 2026 comes at a particularly meaningful moment for me. As I prepare for my graduation thesis and future graduate study, I hope to join VSSS not only to learn from leading scientists, but also to sharpen the way I formulate research questions, think about scientific rigor, and understand the broader responsibility of doing research in an age increasingly shaped by AI and interdisciplinary challenges. I believe the school's environment, where young researchers from different fields gather to question, discuss, and rethink science together, would help me place my engineering research interests within a wider scientific and societal context.

At the same time, I hope to contribute to VSSS from the perspective of an engineering student who has learned across Vietnam, Indonesia, Taiwan, and Singapore, and who is trying to connect robotics, control, AI, and energy systems toward meaningful technological problems. I do not see VSSS merely as a short academic program, but as an opportunity to grow into a more thoughtful, responsible, and intellectually mature young researcher.

I sincerely hope to be part of VSSS 2026 and continue this process of rethinking science, rethinking myself, and preparing for a long-term journey in research.

Sincerely,

Long Nguyen Khac